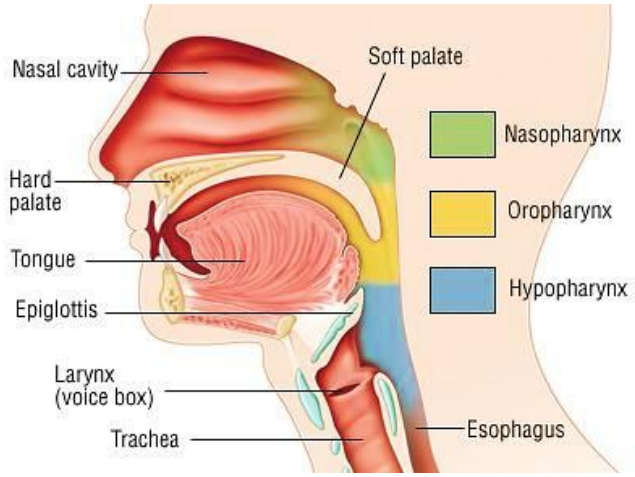
Pharynx, tonsils & oesophagus

The pharyngeal wall and spaces, nasopharynx, oropharynx with the tonsil, hypopharynx, and the cervical oesophagus.

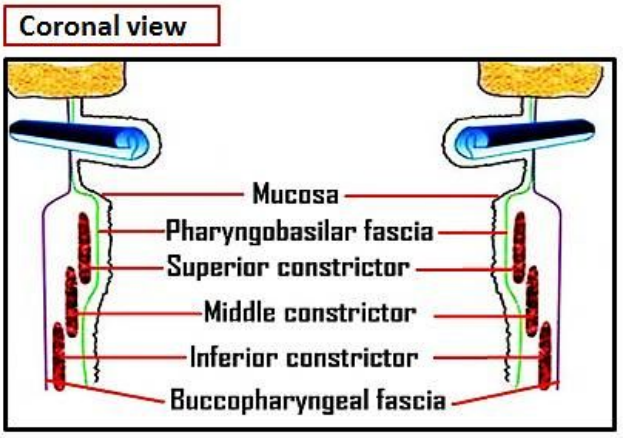
1. General features and wall
2. Muscles of the pharynx
3. Killian's dehiscence and the oesophageal triangles
4. Pharyngeal spaces
5. Waldeyer's ring
6. Nasopharynx
7. Oropharynx
8. Palatine tonsil
9. Hypopharynx
10. Cervical oesophagus
11. In theatre
12. On imaging

01 General features and wall

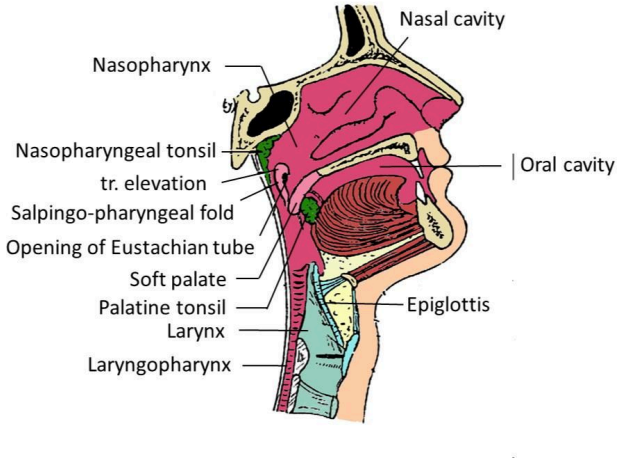
- Fibromuscular tube **12–14 cm** long from the skull base to the **lower border of the cricoid (C6)**, where it becomes the oesophagus. The pharyngo-oesophageal junction is the narrowest part of the gut apart from the appendix.
- **Four layers:** mucosa, **pharyngobasilar fascia** (thick near the skull base), muscle, **buccopharyngeal fascia**.
- Epithelium: ciliated columnar in the nasopharynx, stratified squamous elsewhere.



Divisions of the pharynx
ENT Steps Anatomy, p. 194

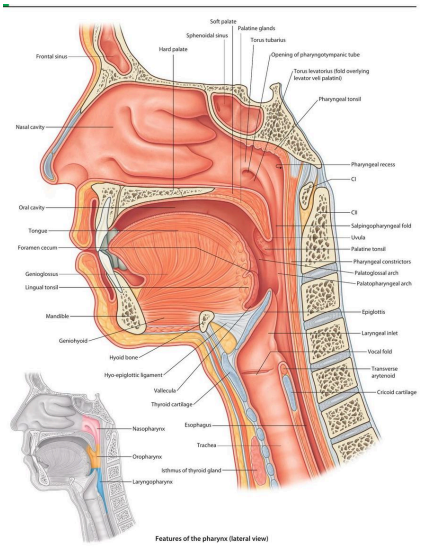


Layers of the pharyngeal wall
ENT Steps Anatomy, p. 197



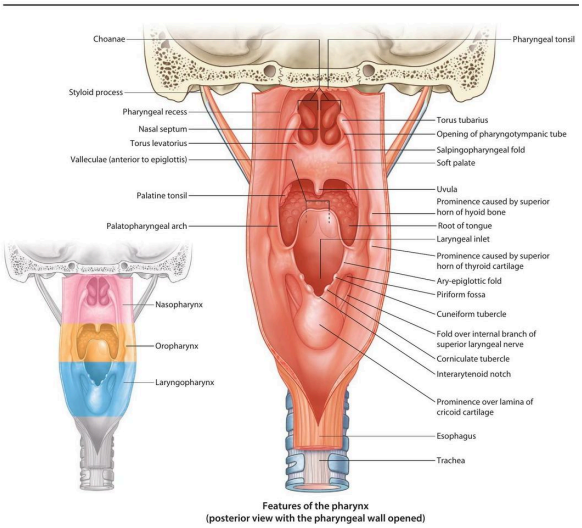
Cavity of the pharynx in sagittal section

Dr. Sameh Doss handout, p. 168



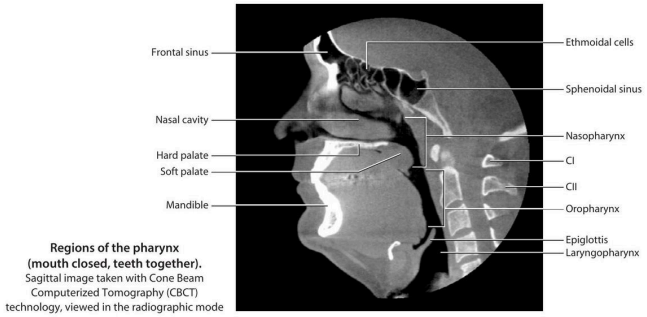
Features of the pharynx (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 552



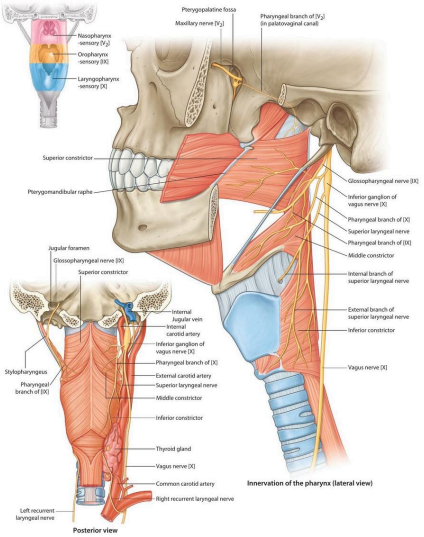
Features of the pharynx (posterior view, pharyngeal wall opened)

Gray's Atlas of Anatomy, 2nd ed., p. 553



Regions of the pharynx

Gray's Atlas of Anatomy, 2nd ed., p. 553

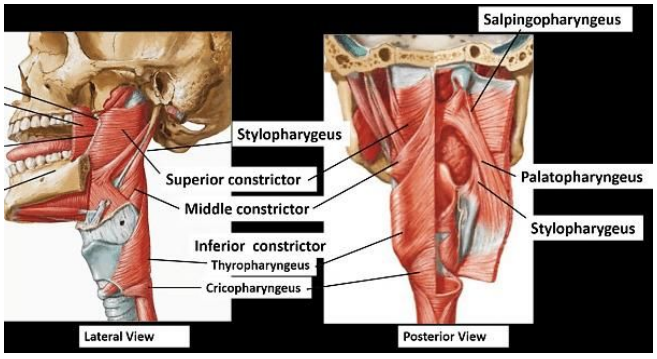


Innervation of the pharynx (lateral and posterior views)

Gray's Atlas of Anatomy, 2nd ed., p. 556

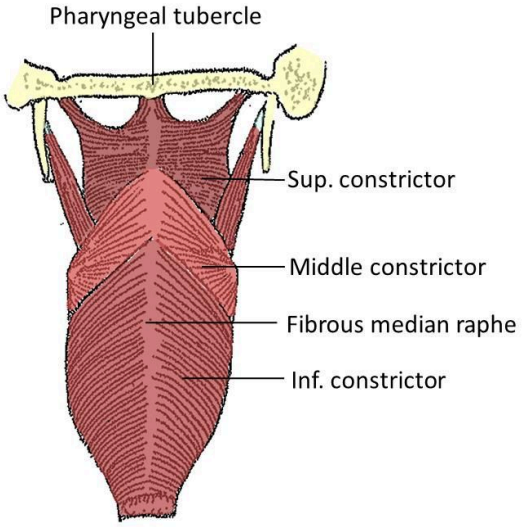
02 Muscles of the pharynx

- **Outer circular layer:** superior, middle and inferior constrictors. The inferior constrictor has **thyropharyngeus** (oblique) and **cricopharyngeus** (transverse, the upper oesophageal sphincter).
- **Inner longitudinal layer:** stylopharyngeus, salpingopharyngeus, palatopharyngeus.
- **Nerve supply:** all by the **pharyngeal plexus (X)** except **stylopharyngeus (IX)**.
- **Gaps:** above the superior constrictor (**sinus of Morgagni**): ET, levator veli palatini, ascending palatine artery. Between superior and middle: stylopharyngeus, IX, stylohyoid ligament. Between middle and inferior: internal laryngeal nerve and superior laryngeal vessels. Below the inferior: recurrent laryngeal nerve and inferior laryngeal vessels.



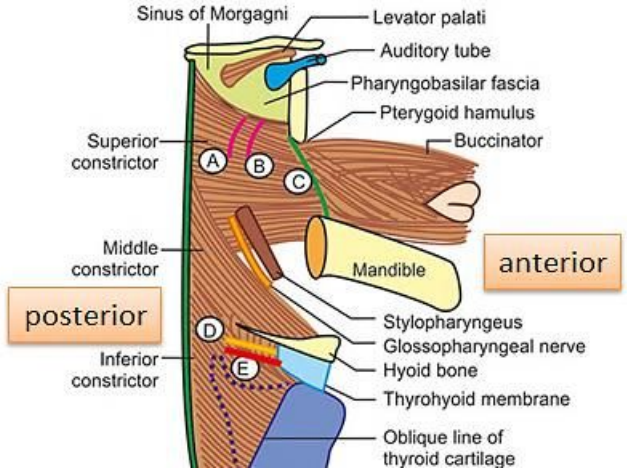
Pharyngeal constrictors: lateral and posterior views

ENT Steps Anatomy, p. 197



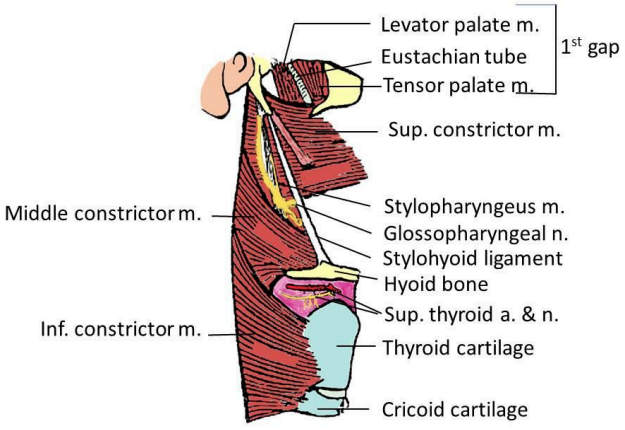
Constrictors of the pharynx, posterior view

Dr. Sameh Doss handout, p. 166



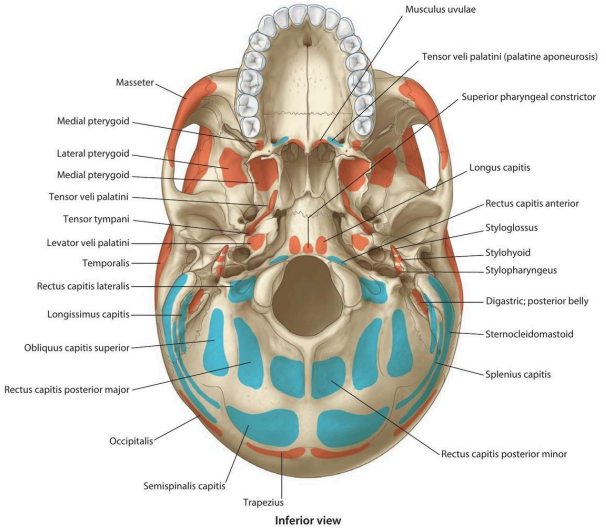
Gaps between the constrictors and the structures passing through them

ENT Steps Anatomy, p. 204



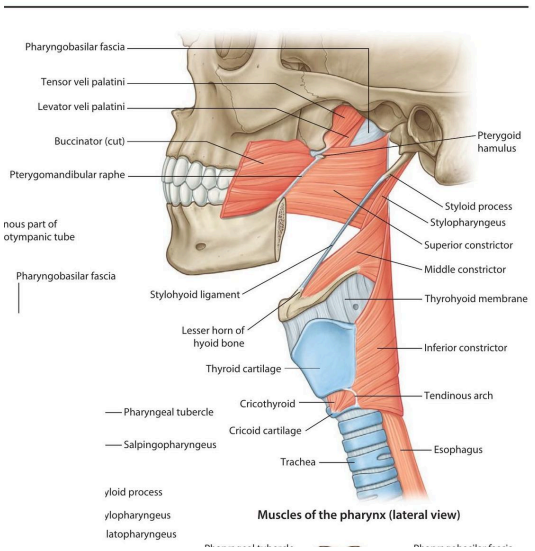
Gaps between the constrictors, lateral view

Dr. Sameh Doss handout, p. 167



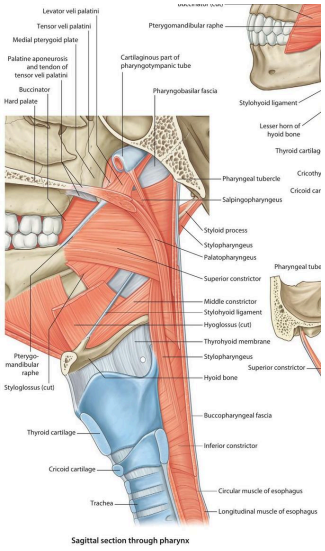
Skull, inferior view: muscle attachments

Gray's Atlas of Anatomy, 2nd ed., p. 489



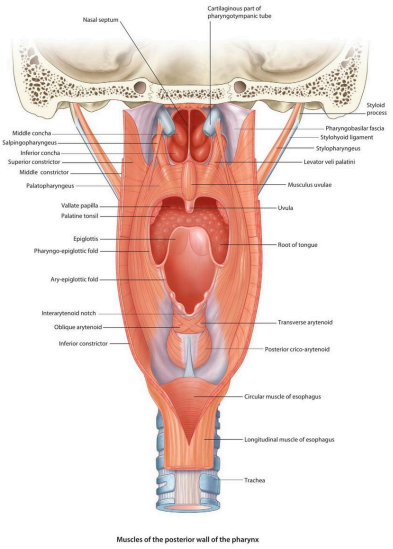
Muscles of the pharynx (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 554



Sagittal section through the pharynx

Gray's Atlas of Anatomy, 2nd ed., p. 554



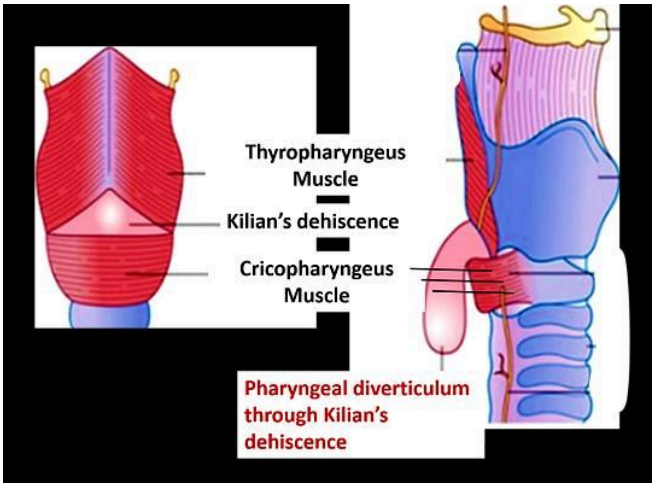
Muscles of the posterior wall of the pharynx

Gray's Atlas of Anatomy, 2nd ed., p. 555

03 Killian's dehiscence and the oesophageal triangles

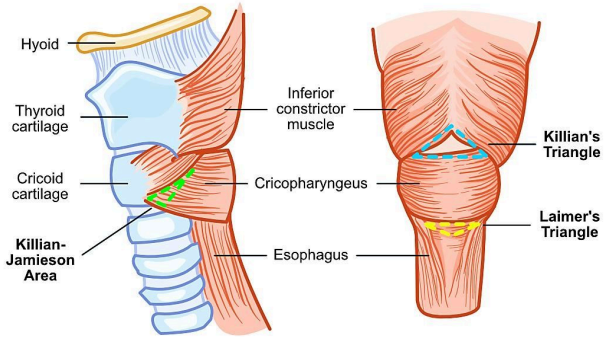
- **Killian's dehiscence:** weak area between thyropharyngeus and cricopharyngeus, posteriorly.
- **Laimer's triangle:** below cricopharyngeus, between it and the oesophageal longitudinal muscle.
- **Killian–Jamieson area:** lateral, below cricopharyngeus.

> **CLINICAL PEARL** Zenker's (pharyngeal pouch) herniates through Killian's dehiscence when cricopharyngeus fails to relax. The same spot is the commonest site of perforation at rigid oesophagoscopy ('gateway of tears').



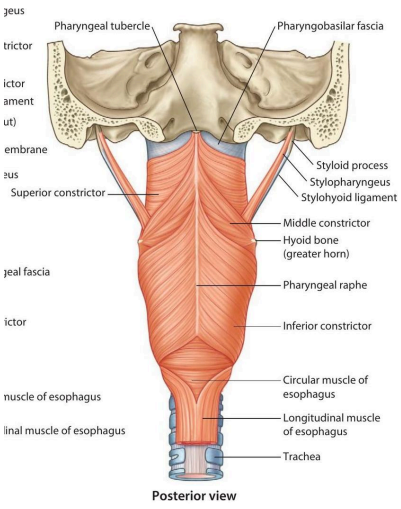
Killian's dehiscence and pharyngeal pouch

ENT Steps Anatomy, p. 198



Killian's and Laimer's triangles

ENT Steps Anatomy, p. 225



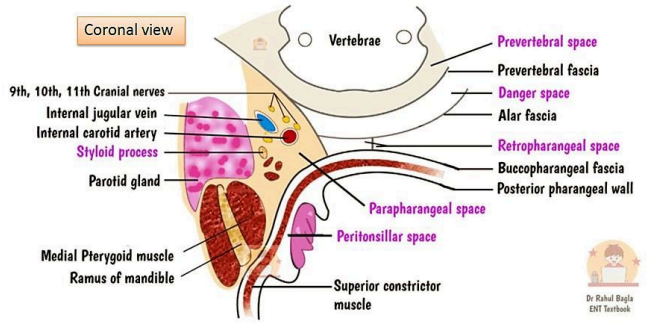
Pharyngeal constrictors (posterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 554

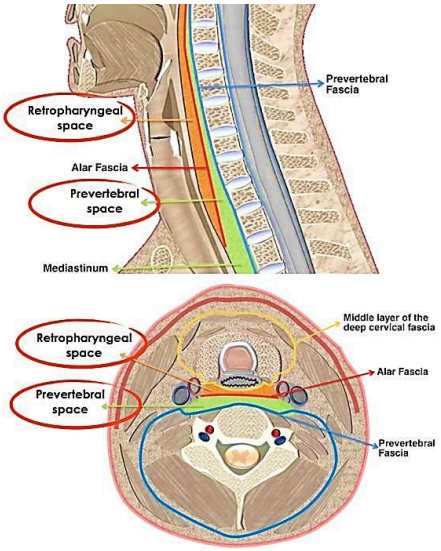
04 Pharyngeal spaces

- **Parapharyngeal space:** inverted pyramid from the skull base to the hyoid. Divided by the styloid and tensor-vascular-styloid fascia into **prestyloid** (deep lobe of parotid, fat, minor salivary glands) and **poststyloid** (ICA, IJV, IX–XII, sympathetic chain, nodes).
- **Retropharyngeal space:** between the buccopharyngeal fascia and the alar fascia, from the skull base to about T2; contains the retropharyngeal nodes (lateral = **Rouvière's** node).
- **Danger space:** between the alar and prevertebral fascia, from the skull base to the diaphragm.

> **CLINICAL PEARL** Prestyloid masses are mostly pleomorphic adenomas; poststyloid masses are mostly paragangliomas and schwannomas. Retropharyngeal nodes regress after about 5 years, so retropharyngeal abscess is a disease of young children.



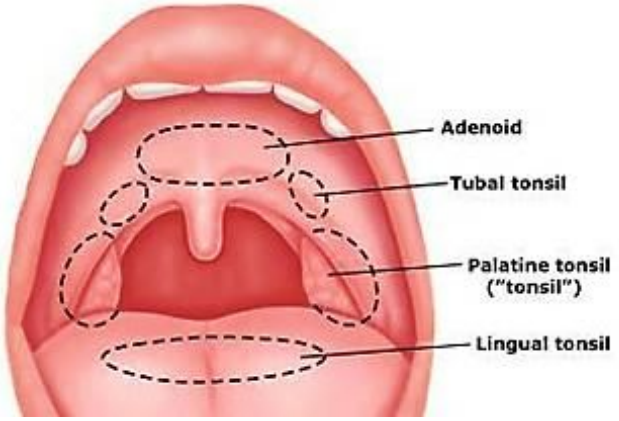
Parapharyngeal and retropharyngeal spaces
ENT Steps Anatomy, p. 198



Retropharyngeal, danger and prevertebral spaces
ENT Steps Anatomy, p. 277

05 Waldeyer's ring

- Ring of mucosa-associated lymphoid tissue at the entrance of the aerodigestive tract: **adenoids, tubal tonsils, lateral pharyngeal bands, palatine tonsils, posterior wall nodules, lingual tonsil**.
- Produce IgA and IgG; crypts increase the surface exposed to antigens. Most active between 4 and 10 years, then involute.
- **Adenoids vs palatine tonsils:** adenoid is single, in the nasopharynx, with folds but no true crypts, **no capsule**, respiratory epithelium, usually gone by adulthood. Tonsils are paired, with crypts, a **capsule** and squamous epithelium.



Waldeyer's ring seen through the mouth
ENT Steps Anatomy, p. 199

TABLE 1 Differences between palatine tonsils and adenoids

	Adenoids	Palatine tonsils
Number	Single	One on each side
Site	Nasopharynx	Tonsillar fossa in oropharynx
Crypts or furrows	Only furrows	Only crypts
Capsule	Absent	Present
Epithelium	Ciliated columnar	Squamous stratified
In adults after 20 years of age	Absent	Present

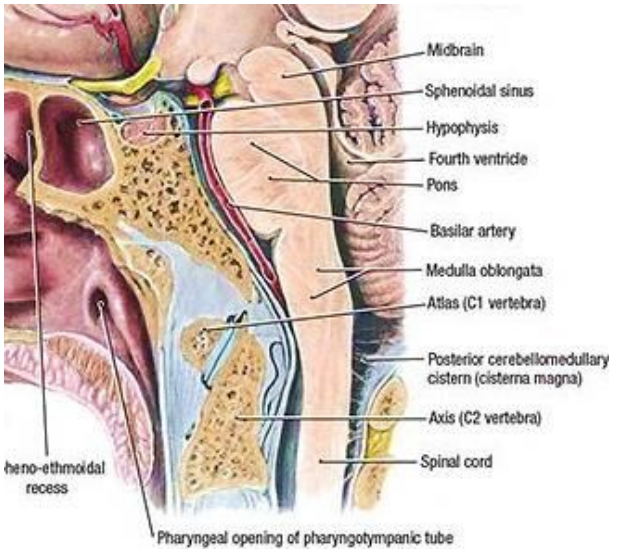
Adenoids versus palatine tonsils
ENT Steps Anatomy, p. 208

06 Nasopharynx

- Behind the choanae, from the skull base to the soft palate (level of the hard palate and atlas).
- **Roof and back wall:** basisphenoid, basiocciput and atlas, with the **adenoids** at their junction (largest up to about 6 years). The **pharyngeal bursa** marks the notochord attachment; **Rathke's pouch** remnant lies above.
- **Lateral wall: ET orifice** about 1.25 cm behind the inferior turbinate, **torus tubarius**, **salpingopharyngeal fold**, tubal tonsil and the **fossa of Rosenmüller** behind and above the torus.

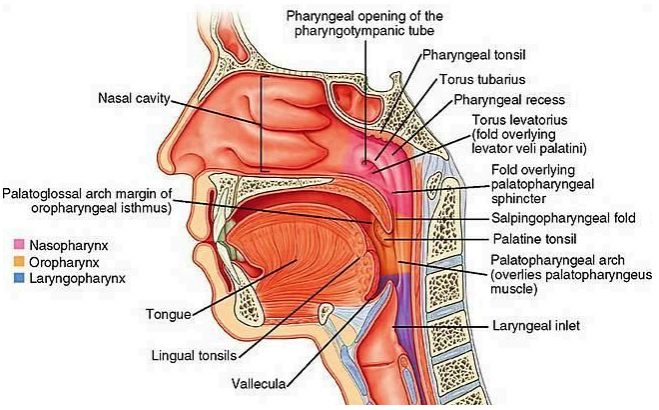
- **Floor:** soft palate; the nasopharyngeal isthmus closes against **Passavant's ridge** (palatopharyngeus) in swallowing and speech.
- **Lymph:** retropharyngeal (Rouvière) and upper deep cervical nodes, spinal accessory chain, **often both sides**.

> **CLINICAL PEARL** The fossa of Rosenmüller is the commonest site of nasopharyngeal carcinoma. A unilateral middle ear effusion in an adult needs a nasopharyngeal look.



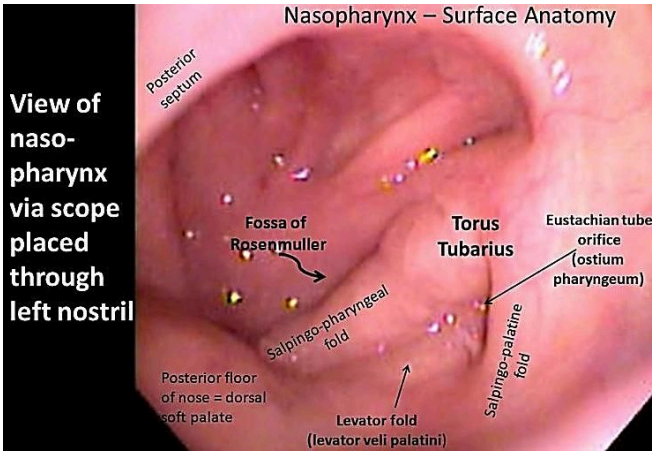
Nasopharynx in sagittal section

ENT Steps Anatomy, p. 200



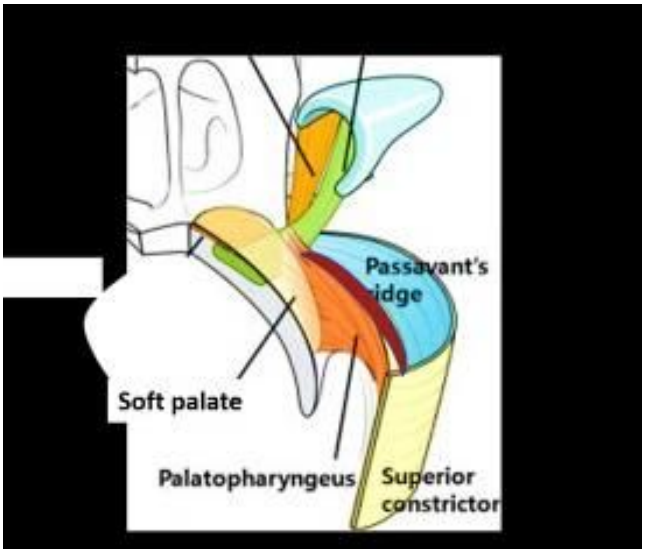
Lateral wall of the pharynx: torus tubarius, fossa of Rosenmuller

ENT Steps Anatomy, p. 203



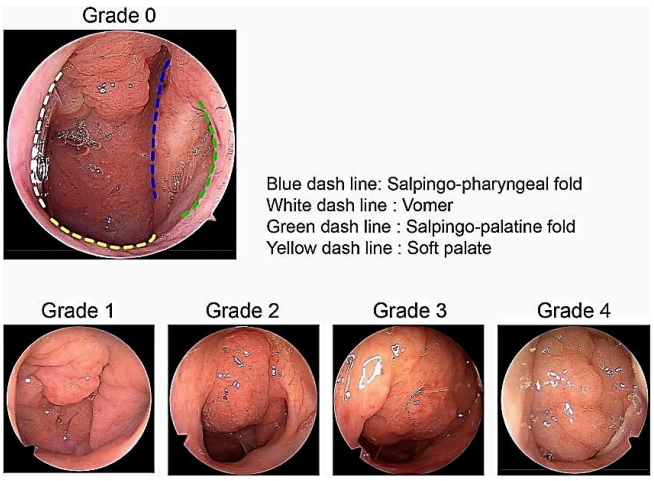
Endoscopic view of the nasopharynx

ENT Steps Anatomy, p. 203



Passavant's ridge

ENT Steps Anatomy, p. 205



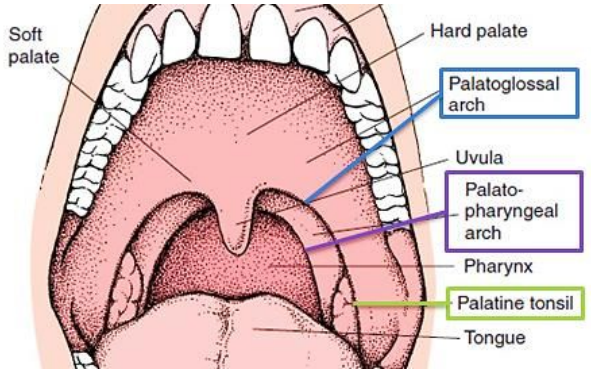
Endoscopic grading of adenoids

ENT Steps Anatomy, p. 207

07 Oropharynx

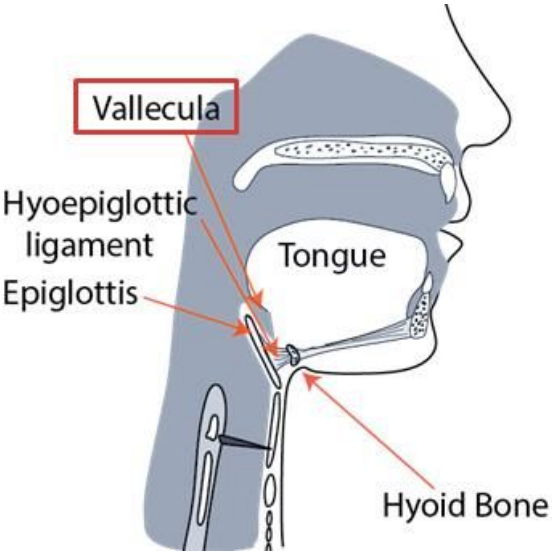
- From the soft palate to the level of the **hyoid (tip of epiglottis)**; opens forward into the mouth at the **oropharyngeal isthmus**.
- **Subsites:** soft palate and uvula, **tonsillar region** (tonsil, pillars), **base of tongue** with lingual tonsil and **valleculae**, posterior wall.
- Valleculae lie between the tongue base and epiglottis, divided by the median glossoepiglottic fold; the lateral glossoepiglottic folds (sometimes called pharyngoepiglottic folds) bound them laterally.
- **Lymph:** upper jugular chain, especially the **jugulodigastric (tonsillar) node**, plus retropharyngeal and parapharyngeal nodes.

> **CLINICAL PEARL** Tongue base lymphatics drain to both sides, so base of tongue cancer often presents with bilateral nodes.



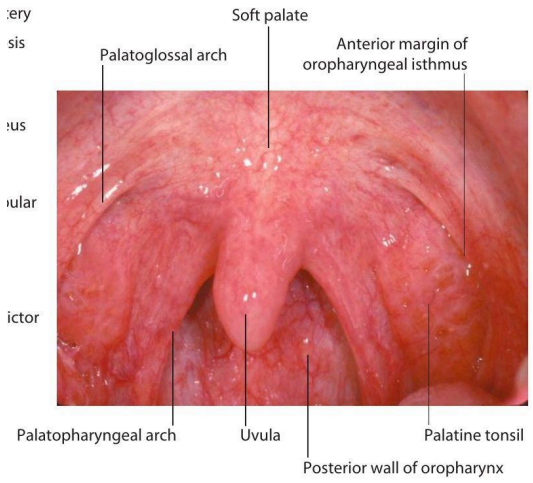
Oropharynx through the open mouth

ENT Steps Anatomy, p. 210



Vallecula and hyoepiglottic ligament

ENT Steps Anatomy, p. 210



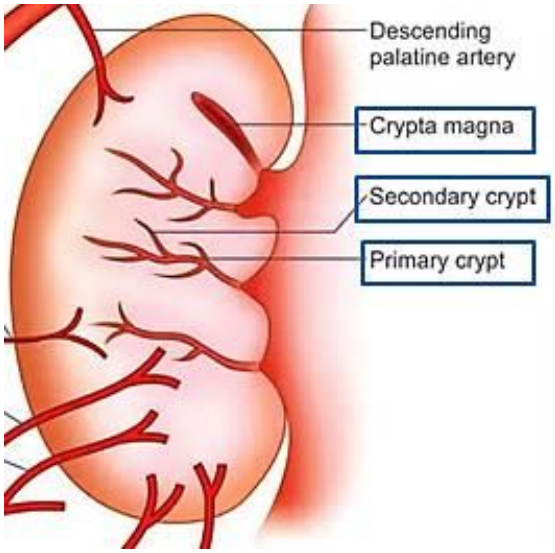
Open mouth: soft palate, arches and palatine tonsils

Gray's Atlas of Anatomy, 2nd ed., p. 582

08 Palatine tonsil

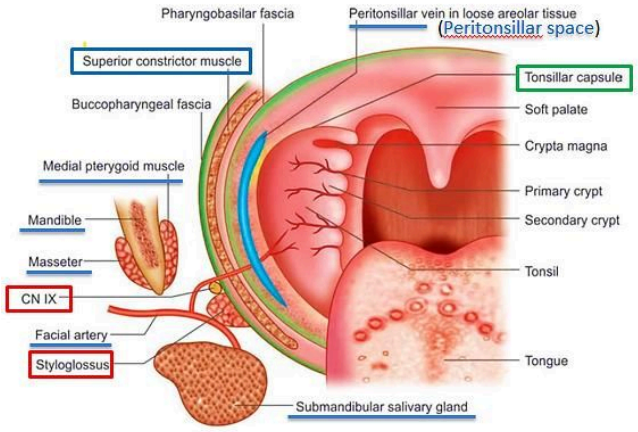
- In the tonsillar fossa between the **palatoglossal (anterior) and palatopharyngeal (posterior) arches**. Supratonsillar fossa above it.
- **12–15 crypts**; the large **crypta magna** (intratonsillar cleft) near the upper pole is a remnant of the 2nd pouch.
- **Capsule** on the lateral surface, separated from the bed by loose areolar tissue: the plane for tonsillectomy and the site of **peritonsillar abscess (quinsy)**.
- **Bed:** superior constrictor, and lateral to it the facial artery, submandibular gland, posterior belly of digastric, medial pterygoid, angle of mandible; **glossopharyngeal nerve** and styloglossus.
- **Arteries (ECA): tonsillar branch of facial (main)**, ascending palatine, ascending pharyngeal, dorsal lingual, descending palatine.
- **Veins:** the **paratonsillar (external palatine) vein** on the lateral surface drains to the common facial vein and pharyngeal plexus.
- **Nerves:** IX and lesser palatine nerves (V2). **Lymph:** jugulodigastric node.

> **CLINICAL PEARL** Primary bleeding at tonsillectomy is usually venous from the paratonsillar vein; secondary bleeding (days 5–10) follows infection of the tonsillar fossa. Post-operative ear pain is referred via IX. The ICA lies about 2.5 cm behind and lateral to the tonsil.



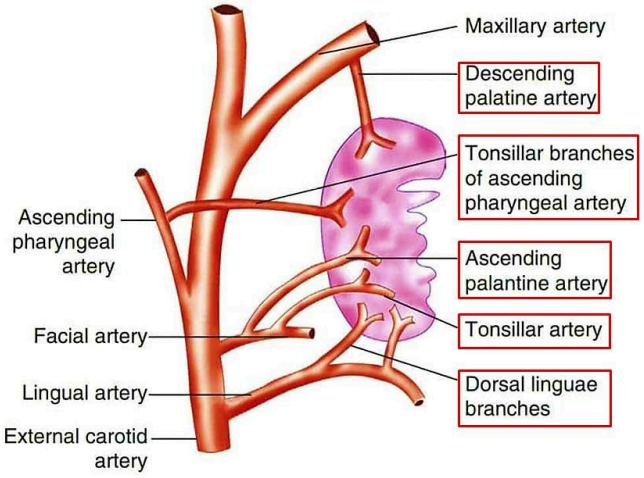
Tonsillar crypts and blood vessels

ENT Steps Anatomy, p. 214



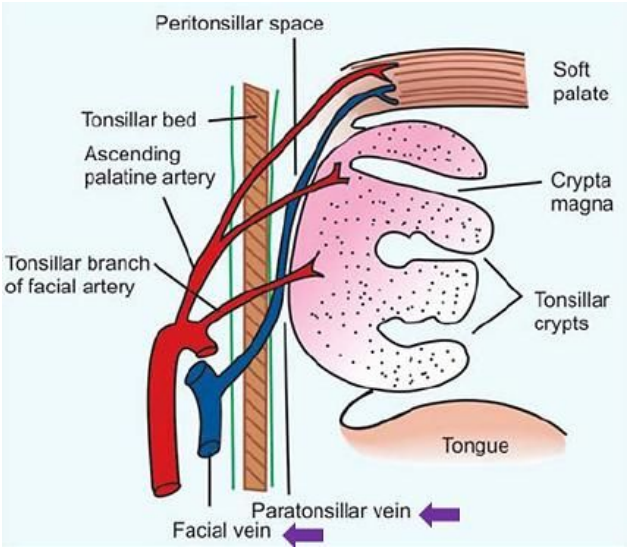
Bed of the tonsil in horizontal section

ENT Steps Anatomy, p. 215



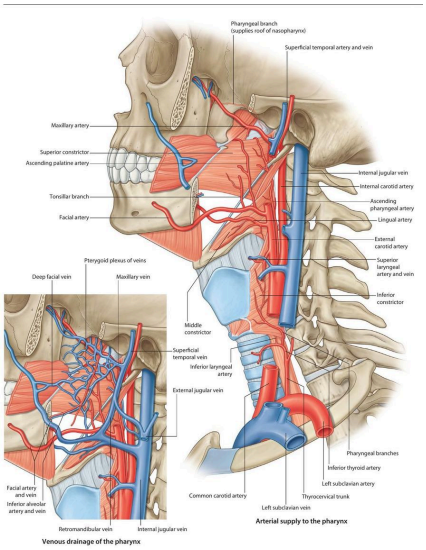
Arterial supply of the palatine tonsil

ENT Steps Anatomy, p. 217



Paratonsillar vein and tonsillar arteries

ENT Steps Anatomy, p. 218



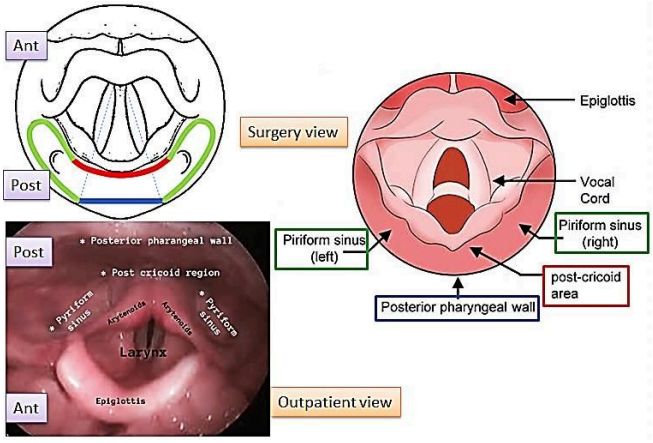
Arterial supply and venous drainage of the pharynx

Gray's Atlas of Anatomy, 2nd ed., p. 557

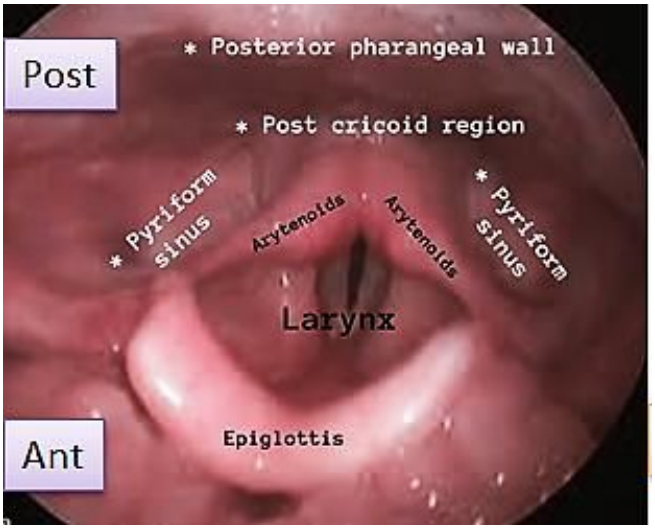
09 Hypopharynx

- From the hyoid to the **lower border of the cricoid (C3–C6)**, behind and beside the larynx.
- **Three subsites: piriform sinus**, **postcricoid region**, **posterior pharyngeal wall**.
- **Piriform sinus:** lateral = thyrohyoid membrane and thyroid lamina; medial = aryepiglottic fold, arytenoid and cricoid; above = pharyngoepiglottic fold. Foreign bodies lodge here.
- The **internal laryngeal nerve** runs under the mucosa of the piriform sinus (local anaesthesia site).
- **Lymph:** piriform sinus through the thyrohyoid membrane to the upper jugular chain; postcricoid to paratracheal and supraclavicular nodes.

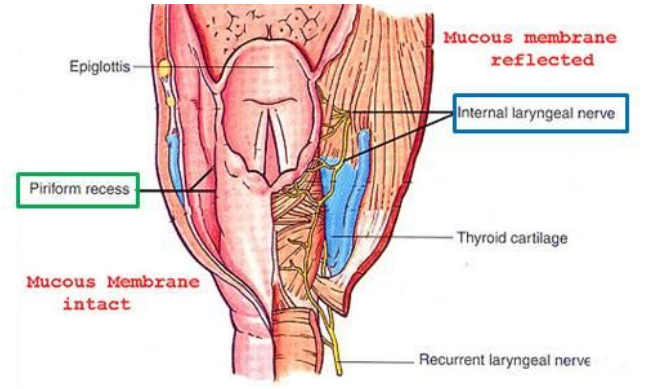
> **CLINICAL PEARL** Piriform sinus cancer causes referred otalgia through the internal laryngeal nerve (X). Postcricoid carcinoma is associated with Plummer–Vinson syndrome in women.



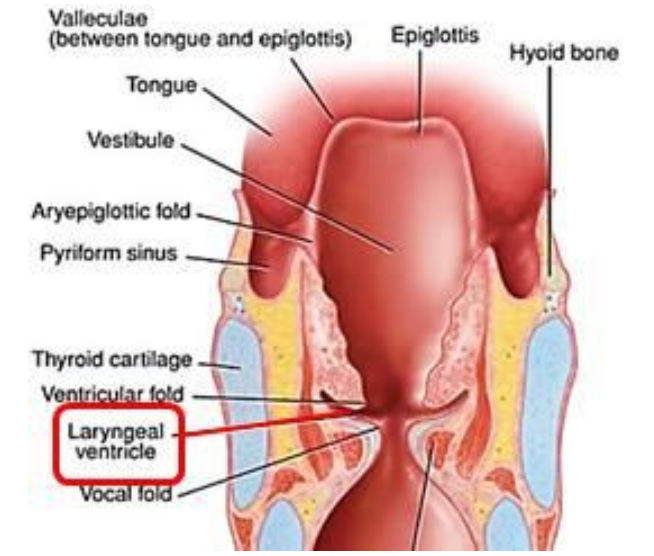
Three subsites of the hypopharynx
ENT Steps Anatomy, p. 220



Endoscopic view of the hypopharynx and larynx
ENT Steps Anatomy, p. 221



Internal laryngeal nerve under the mucosa of the piriform fossa
ENT Steps Anatomy, p. 221



Vallecula, piriform sinus and laryngeal ventricle
ENT Steps Anatomy, p. 240

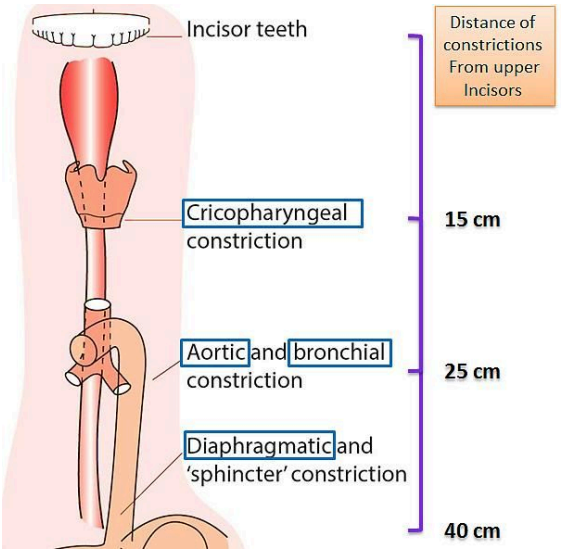
10 Cervical oesophagus

- About **25 cm** from the cricoid (C6) to the cardia (T11); about **40 cm** from the upper incisors.

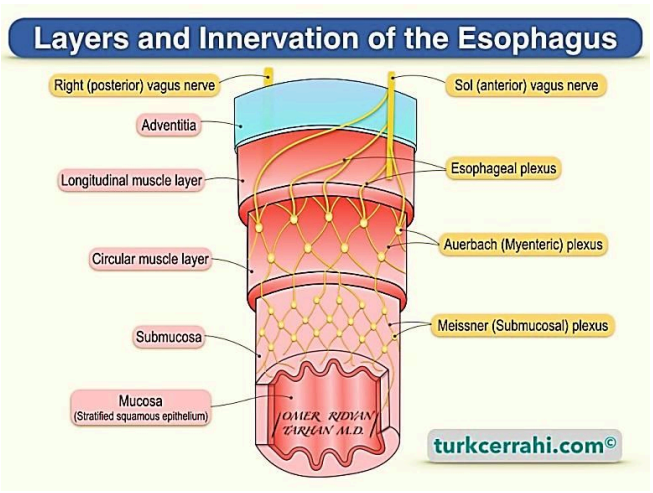
- **Constrictions** (from the upper incisors): cricopharyngeus **15 cm** (C6, the narrowest), aortic arch **22 cm** (T4), left main bronchus **25 cm** (T5), diaphragm **40 cm** (T10).
- Layers: squamous mucosa, submucosa, inner circular and outer longitudinal muscle, adventitia (no serosa). Upper third striated, lower third smooth.
- **Upper oesophageal sphincter:** a high-pressure zone about 2.5–4.5 cm long (cricopharyngeus, about 1 cm wide, plus adjacent muscle); closed at rest.
- **Lymph:** cervical part to deep cervical and paratracheal nodes.

> **CLINICAL PEARL** Swallowed coins in children most often stick at cricopharyngeus.

> **SOURCES DIFFER CONSTRICTIONS:** ENT Steps 15 / 22 / 25 / 40 cm (C6 / T4 / T5 / T10). Dhingra 15 / 22.5 / 27.5 / 37.5–40 cm, with the bronchial constriction at about T6.



Constrictions of the oesophagus and their distance from the incisors
ENT Steps Anatomy, p. 226

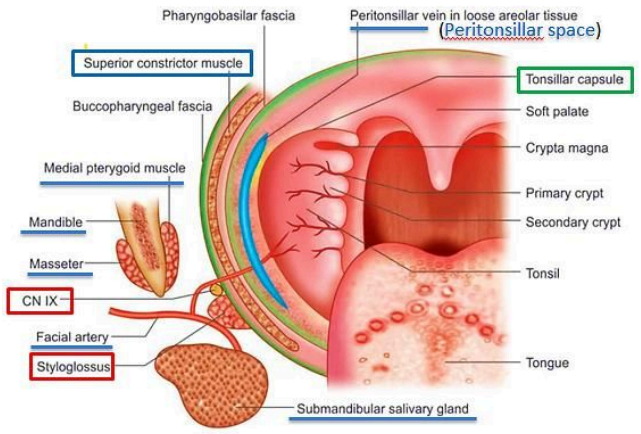


Layers and innervation of the oesophagus
ENT Steps Anatomy, p. 227

11 Surgical anatomy: In theatre

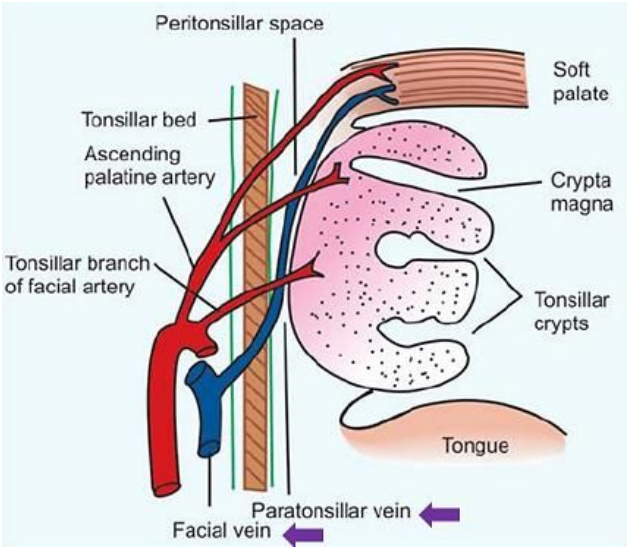
- **Tonsillectomy:** grasp the tonsil, incise the anterior pillar mucosa, find the capsule and stay on it; the paratonsillar vein runs in the bed; the lower pole connects to the lingual tonsil.
- **Adenoidectomy:** check for a submucous cleft first; avoid the torus tubarius laterally (stenosis of the ET opening).
- **Quinsy drainage:** incise at the point of maximal bulge, or where a horizontal line from the base of the uvula meets a vertical line from the anterior pillar; avoid going lateral (ICA).
- **Rigid oesophagoscopy:** risk of perforation at Killian's dehiscence; neck extension and a well-lubricated scope.
- **Endoscopic stapling of a pharyngeal pouch** divides the party wall (cricopharyngeus).

> **CLINICAL PEARL** A pulsating peritonsillar swelling may be an aberrant ICA or aneurysm; aspirate before incising.



Bed of the tonsil in horizontal section

ENT Steps Anatomy, p. 215

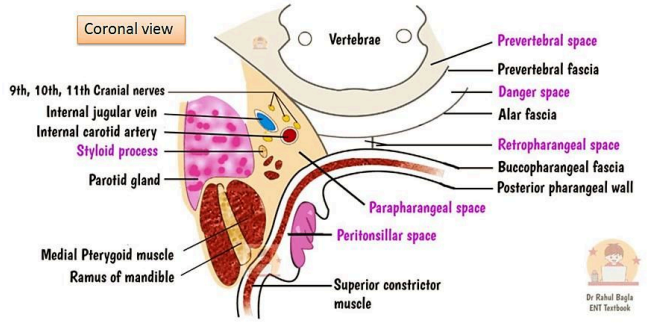


Paratonsillar vein and tonsillar arteries

ENT Steps Anatomy, p. 218

12 Radiological anatomy: On imaging

- **Lateral neck X-ray:** prevertebral soft tissue at C2 normally under about 7 mm, at C6 under about 22 mm in adults (14 mm in children).
- **CT with contrast** for deep neck abscess: rim-enhancing collection; distinguishes abscess from phlegmon.
- **Parapharyngeal tumours:** prestyloid lesions push the fat medially and the ICA backward; poststyloid lesions push the ICA forward.
- **Barium swallow** shows a pouch at the back of the pharyngo-oesophageal junction and webs (Plummer–Vinson).



Parapharyngeal and retropharyngeal spaces

ENT Steps Anatomy, p. 198

T Tables

Gaps in the pharyngeal wall

Gap	Structures passing
Skull base – superior constrictor (sinus of Morgagni)	Eustachian tube, levator veli palatini, ascending palatine artery
Superior – middle constrictor	Stylopharyngeus, glossopharyngeal nerve, stylohyoid ligament
Middle – inferior constrictor	Internal laryngeal nerve, superior laryngeal vessels
Below the inferior constrictor	Recurrent laryngeal nerve, inferior laryngeal vessels

Palatine tonsils vs adenoids

	Palatine tonsil	Adenoid
Number	Paired	Single
Site	Tonsillar fossa (oropharynx)	Roof/posterior wall of nasopharynx
Surface	12–15 crypts, crypta magna	Folds, no true crypts
Capsule	Present	Absent
Epithelium	Stratified squamous	Ciliated columnar
In adults	Present	Usually involuted
Main artery	Tonsillar branch of facial	Ascending pharyngeal, ascending palatine

Hypopharynx subsites compared

Subsite	Boundaries	Lymph drainage	Notes
Piriform sinus	Pharyngoepiglottic fold to the oesophageal inlet, beside the larynx	Upper jugular via thyrohyoid membrane	Commonest site (~65–75%); referred otalgia via X
Postcricoid region	Arytenoids to lower cricoid, front wall	Paratracheal, lower jugular	Plummer–Vinson, women
Posterior pharyngeal wall	Hyoid to cricoid level, behind	Retropharyngeal and jugular	Often presents late

Deep neck spaces

Space	Extent	Key contents / source
Retropharyngeal	Skull base to T2	Retropharyngeal nodes; children after URTI
Danger	Skull base to diaphragm	Loose fat; spread to mediastinum
Prevertebral	Skull base to coccyx	Vertebral (TB, osteomyelitis)
Parapharyngeal	Skull base to hyoid	Pre- and poststyloid compartments; dental, tonsil
Submandibular / sublingual	Floor of mouth above and below mylohyoid	Ludwig's angina (2nd and 3rd molars)
Peritonsillar	Tonsil capsule to superior constrictor	Quinsy

Arteries of the palatine tonsil

Artery	Parent	Pole
Tonsillar branch (main)	Facial	Lower pole
Ascending palatine	Facial	Lower pole
Dorsal lingual	Lingual	Lower pole
Ascending pharyngeal	ECA	Upper pole
Lesser (descending) palatine	Maxillary	Upper pole

Veins: paratonsillar (external palatine) → common facial vein and pharyngeal plexus.

Pharyngeal constrictors

Muscle	Posterior attachment	Anterior attachment	Nerve	Action
Superior constrictor	Pharyngeal raphe	Pterygomandibular raphe, pterygoid hamulus and adjacent mandible	Vagus (pharyngeal plexus)	Constricts pharynx
Middle constrictor	Pharyngeal raphe	Greater and lesser horns of hyoid, stylohyoid ligament	Vagus (pharyngeal plexus)	Constricts pharynx
Inferior constrictor	Pharyngeal raphe	Oblique line of thyroid cartilage, cricoid, and the band between them over cricothyroid	Vagus (pharyngeal plexus); cricopharyngeus also RLN	Constricts pharynx; cricopharyngeus is the upper oesophageal sphincter

Gray's Atlas, constrictor table. The three overlap like stacked cups, open anteriorly.

Longitudinal muscles of the pharynx

Muscle	Origin	Insertion	Nerve	Action
Stylopharyngeus	Medial side of base of styloid process	Pharyngeal wall (passes between superior and middle constrictors)	Glossopharyngeal (IX) — its only muscle	Elevates pharynx
Salpingopharyngeus	Cartilaginous end of the Eustachian tube	Pharyngeal wall	Vagus (pharyngeal plexus)	Elevates pharynx
Palatopharyngeus	Upper surface of palatine aponeurosis	Pharyngeal wall and thyroid cartilage	Vagus (pharyngeal plexus)	Elevates pharynx; closes oropharyngeal isthmus

Gray's Atlas. All except stylopharyngeus are supplied by X through the pharyngeal plexus.

Muscles of the soft palate

Muscle	Origin	Insertion	Nerve	Action
Tensor veli palatini	Scaphoid fossa, spine of sphenoid, cartilaginous Eustachian tube	Palatine aponeurosis (tendon hooks round the hamulus)	V3 (via nerve to medial pterygoid)	Tenses palate; opens the Eustachian tube
Levator veli palatini	Petrous temporal bone in front of carotid canal; tube cartilage	Upper surface of palatine aponeurosis	Vagus (pharyngeal plexus)	Only elevator of the soft palate above neutral
Palatopharyngeus	Upper surface of palatine aponeurosis	Pharyngeal wall	Vagus (pharyngeal plexus)	Depresses palate; narrows isthmus; elevates pharynx
Palatoglossus	Under surface of palatine aponeurosis	Side of tongue	Vagus (pharyngeal plexus)	Depresses palate; elevates back of tongue
Musculus uvulae	Posterior nasal spine	Mucosa of uvula	Vagus (pharyngeal plexus)	Elevates and retracts uvula

Gray's Atlas. Tensor is the only palatal muscle not supplied by the pharyngeal plexus.

M Mnemonics

Oesophageal constrictions

15, 22, 25, 40

Cricopharyngeus, aortic arch, left main bronchus, diaphragm (cm from incisors)

Tonsil arteries

Facial is the main road

Tonsillar branch of facial is the main artery; plus ascending palatine, ascending pharyngeal, dorsal lingual, descending palatine

Waldeyer's ring

Adenoids Take Lessons Playing Leading Notes

Adenoid, Tubal tonsils, Lateral bands, Palatine tonsils, Lingual tonsil, Nodules of posterior wall

Constrictor gaps

Morgagni, Stylopharyngeus, Internal laryngeal, Recurrent

Above superior; superior–middle; middle–inferior; below inferior

Hypopharynx subsites

PPP

Piriform sinus, Postcricoid, Posterior pharyngeal wall

Q Rapid recall

1. **Which pharyngeal muscle is not supplied by the vagus?** Stylopharyngeus (IX).
2. **Where is Killian's dehiscence?** Between thyropharyngeus and cricopharyngeus, posteriorly.
3. **Structures passing through the sinus of Morgagni?** Eustachian tube, levator veli palatini, ascending palatine artery.
4. **Commonest site of nasopharyngeal carcinoma?** Fossa of Rosenmüller.
5. **Main artery of the palatine tonsil?** Tonsillar branch of the facial artery.
6. **Usual source of primary bleeding at tonsillectomy?** The paratonsillar (external palatine) vein. Secondary bleeding (days 5–10) is due to infection of the fossa.
7. **Three subsites of the hypopharynx?** Piriform sinus, postcricoid region, posterior pharyngeal wall.
8. **Constrictions of the oesophagus and distances from the incisors?** Cricopharyngeus 15 cm, aortic arch 22 cm, left main bronchus 25 cm, diaphragm 40 cm.
9. **Contents of the poststyloid parapharyngeal space?** ICA, IJV, IX–XII, sympathetic chain and lymph nodes.
10. **Length of the pharynx?** About 12–14 cm, from the skull base to the lower border of the cricoid (C6).
11. **Four layers of the pharyngeal wall?** Mucosa, pharyngobasilar fascia, muscle, buccopharyngeal fascia.
12. **Two parts of the inferior constrictor?** Thyropharyngeus and cricopharyngeus.
13. **What raises Passavant's ridge?** Palatopharyngeus (upper fibres of superior constrictor).
14. **What is Rouvière's node?** The highest lateral retropharyngeal node, at the level of the atlas.
15. **Extent of the danger space?** Skull base to the diaphragm.
16. **What forms the bed of the tonsil?** Superior constrictor, with the glossopharyngeal nerve, styloglossus and facial artery beyond.
17. **Lymph drainage of the palatine tonsil?** Jugulodigastric node.
18. **What is the crypta magna?** Large intratonsillar cleft near the upper pole, remnant of the 2nd pouch.
19. **Boundaries of the piriform sinus?** Lateral: thyrohyoid membrane and thyroid lamina. Medial: aryepiglottic fold, arytenoid, cricoid. Above: pharyngoepiglottic fold.
20. **Postcricoid carcinoma association?** Plummer–Vinson syndrome, mainly in women.
21. **Length of the upper oesophageal sphincter high-pressure zone?** About 2.5–4.5 cm (cricopharyngeus itself is about 1 cm wide).

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.